INNOVATION MATRIARCH

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CHANDRADEEP KUMAR

THE YOUNGEST OF THREE siblings and the only daughter of a schoolteacher, Nallathamby Kalaiselvi remembers having a brush with gender discrimination umpteen times while growing up in rural Ambasamudram in southern Tamil Nadu. But having supportive parents helped. After she completed her M.Sc. in chemistry from a government college in Coimbatore, her parents encouraged her to pursue a Ph.D.—not a common thing in those days.

But that was not her calling. Kalaiselvi instead opted to pursue her passion in electrochemistry to develop lithium-ion (Li-ion) batteries—the ones that power our mobile phones and electric cars today—when the technology was still at a nascent stage. It was a study tour to the Council for Scientific & Industrial Research (CSIR)'s Central Electrochemical Research Institute (CECRI) in Karaikudi in 1989 that had motivated her to join it as a junior scientist in 1997. "Lithium was hardly available," she recalls, "and we faced several challenges. The coin cell components were expensive and hard to come by as they had to be imported. Moreover, the cell fabrication-related infrastructure was limited, and we faced challenges in trial experiments and cell assembly. These very challenges egged me on to innovate."

In her 25-year stint at the CSIR-CECRI, Kalaiselvi worked extensively on developing new electrode materials to improve the storage capacity of Li-ion batteries, even as her leadership skills and administrative acumen saw her being appointed as its director. It was in August 2022 that Kalaiselvi was appointed as the director-general of the CSIR, the first woman at the helm since its foundation in 1942. And thus she has come to steer the national network of 38 premier research institutions, spearheading multiple projects for technological advancements aimed at meeting the United Nation's Sustainable Development Goals (SDGs). Among them, reduction in greenhouse gas emissions and India's carbon footprint, improving access to clean water, sanitation and healthcare, and most importantly working for the well-being of all citizens.

"Today, we have developed green technologies that add value to the robust supply chain to meet the country's future requirements," says Kalaiselvi. "This is in line with the plans laid out for Amrit Kaal—India's vision for 2047." As a first step towards achieving this 'Amrit Kaal', India needs to realise the SDGs by 2030. And being the fountainhead of research in various fields of science and technology in India, the CSIR, led by Kalaiselvi, will be playing a critical role in this mission. ■

–Amarnath K. Menon

ACCOMPLISHMENT

First woman at the helm of CSIR in its 80-year-long history, she steers the national network of 38 premier research institutions

TRUE GRIT

"Believe in yourself and give your best to strive for excellence"

LOOKS UP TO

Dr A.P.J. Abdul Kalam. Can relate to his humble origins and rise through hard work